

Database Technologies – Assignment 7

Q1) Write a query that lists each order number followed by the name of the customer who made the order.

```
w1_99099_vaibhavi>SELECT orders.onum, customers.cname
-> FROM orders
-> INNER JOIN customers
-> ON orders.cnum = customers.cnum;
```

onum	cname
3001	Cisneros
3003	Hoffman
3002	Pereira
3005	Liu
3006	Cisneros
3009	Giovanni
3007	Grass
3008	Clemens
3010	Grass
3011	Clemens

10 rows in set (0.00 sec)

```
w1_99099_vaibhavi>
```

Q2) Write a query that gives the names of both the salesperson and the customer for each order along with the order number.

```
w1_99099_vaibhavi>SELECT orders.onum, customers.cname, salespeople.sname
-> FROM orders
-> INNER JOIN customers ON orders.cnum = customers.cnum
-> INNER JOIN salespeople ON orders.snum = salespeople.snum;
```

onum	cname	sname
3003	Hoffman	Peel
3009	Giovanni	Axelrod
3005	Liu	Serres
3010	Grass	Serres
3007	Grass	Serres
3011	Clemens	Peel
3008	Clemens	Peel
3006	Cisneros	Rifkin
3001	Cisneros	Rifkin
3002	Pereira	Motika

10 rows in set (0.00 sec)

```
w1_99099_vaibhavi>
```

Q3) Write a query that produces all customers serviced by salespeople with a commission above 12%. Display the customer's name, the salesperson's name, and the salesperson's rate of commission

```
w1_99099_vaibhavi>SELECT customers.cname, salespeople.sname, salespeople.com
m
  -> FROM customers
  -> INNER JOIN salespeople ON customers.snum = salespeople.snum
  -> WHERE salespeople.comm > 0.12;
+-----+-----+-----+
| cname | sname | comm |
+-----+-----+-----+
| Liu   | Serres | 0.13 |
| Grass | Serres | 0.13 |
| Cisneros | Rifkin | 0.15 |
+-----+-----+-----+
3 rows in set (0.00 sec)

w1_99099_vaibhavi>
```

Q4) Write a query that calculates the commission earned by each salesperson for orders placed by customers with a rating above 100.

```
ERROR 1054 (42S22): Unknown column 'customer.cnum' in 'on clause'
w1_99099_vaibhavi>SELECT orders.onum, salespeople.comm, (salespeople.comm*or
ders.amt)
  -> FROM orders
  -> INNER JOIN customers ON orders.cnum = customers.cnum
  -> INNER JOIN salespeople ON orders.snum = salespeople.snum
  -> WHERE customers.rating > 100;
+-----+-----+-----+
| onum | comm | (salespeople.comm*orders.amt) |
+-----+-----+-----+
| 3010 | 0.13 | 40.2935 |
| 3007 | 0.13 | 9.8475 |
| 3005 | 0.13 | 670.8585 |
| 3006 | 0.15 | 164.7240 |
| 3001 | 0.15 | 2.8035 |
| 3009 | 0.10 | 171.3230 |
+-----+-----+-----+
6 rows in set (0.00 sec)

w1_99099_vaibhavi>
```

Q5) Write a query that produces all pairs of salespeople who are living in the same city. Exclude combinations of salespeople with themselves as well as duplicate rows with the order reversed.

```
w1_99099_vaibhavi>select s.sname, sum(sp.qty)
-> FROM s
-> JOIN sp ON s.`S#` = sp.`S#`
-> GROUP BY s.SNAME
-> ORDER BY SUM(sp.QTY) DESC
-> LIMIT 1;
```

```
+-----+-----+
| sname | sum(sp.qty) |
+-----+-----+
| Jones |          3200 |
+-----+-----+
1 row in set (0.00 sec)
```

```
w1_99099_vaibhavi>|
```

```
w1_99099_vaibhavi>SELECT a.sname, b.sname, a.city
-> FROM salespeople a, salespeople b
-> WHERE a.city = b.city AND a.sname < b.sname;
```

```
+-----+-----+-----+
| sname | sname | city  |
+-----+-----+-----+
| Motika | Peel  | London |
+-----+-----+-----+
1 row in set (0.00 sec)
```

```
w1_99099_vaibhavi>|
```

PART 2

Q1) Display the Supplier name and the Quantity sold.

```
w1_99099_vaibhavi>SELECT sname, SUM(SP.QTY)
-> FROM S
-> INNER JOIN SP ON s.`S#` = sp.`S#`
-> GROUP BY sname;
```

```
+-----+-----+
| sname | SUM(SP.QTY) |
+-----+-----+
| Smith |          900 |
| Jones |         3200 |
| Blake |          700 |
| Clark |          600 |
| Adams |         3100 |
+-----+-----+
5 rows in set (0.03 sec)
```

```
w1_99099_vaibhavi>2|
```

Q2) Display the Part name and Quantity sold

```
w1_99099_vaibhavi>select pname, sum(sp.qty)
-> FROM P
-> INNER JOIN SP ON p.`P#` = sp.`p#`
-> GROUP BY pname;
+-----+-----+
| pname | sum(sp.qty) |
+-----+-----+
| Nut   |          1000 |
| Screw |          4800 |
| Cam   |          1100 |
| Cog   |          1300 |
| Bolt  |           300 |
+-----+-----+
5 rows in set (0.00 sec)

w1_99099_vaibhavi>|
```

Q3) Display the Job name and Quantity sold.

```
w1_99099_vaibhavi>SELECT Jname, sum(sp.qty)
-> FROM J
-> INNER JOIN SP ON j.`J#` = sp.`J#`
-> GROUP BY jname;
+-----+-----+
| Jname   | sum(sp.qty) |
+-----+-----+
| Sorter  |           800 |
| Console |          3300 |
| Punch   |          1200 |
| Reader  |           500 |
| Collator |          1100 |
| Terminal |           400 |
| Tape    |          1200 |
+-----+-----+
7 rows in set (0.00 sec)

w1_99099_vaibhavi>|
```

Q4) Display the Supplier name, Part name, Job name and Quantity sold.

```
w1_99099_vaibhavi>SELECT sname, pname, jname, qty
-> FROM sp
-> INNER JOIN s
-> ON sp.`S#` = s.`S#`
-> INNER JOIN p
-> ON sp.`P#` = p.`P#`
-> INNER JOIN j
-> ON sp.`J#` = j.`J#`;
```

sname	pname	jname	qty
Smith	Nut	Sorter	200
Smith	Nut	Console	700
Jones	Screw	Sorter	400
Jones	Screw	Punch	200
Jones	Screw	Reader	200
Jones	Screw	Console	500
Jones	Screw	Collator	600
Jones	Screw	Terminal	400
Jones	Screw	Tape	800
Jones	Cam	Punch	100
Blake	Screw	Sorter	200
Blake	Screw	Punch	500
Clark	Cog	Reader	300
Clark	Cog	Tape	300
Adams	Nut	Console	100
Adams	Bolt	Punch	200
Adams	Bolt	Console	100
Adams	Screw	Console	200
Adams	Screw	Console	800
Adams	Cam	Console	400
Adams	Cam	Collator	500
Adams	Cam	Tape	100
Adams	Cog	Punch	200
Adams	Cog	Console	500

24 rows in set (0.00 sec)

Q5) Display the Supplier name, Supplying Parts to a Job in the same City.

```
w1_99099_vaibhavi>SELECT DISTINCT SNAME FROM S
-> INNER JOIN SP
-> ON s.`S#` = sp.`S#`
-> INNER JOIN J on J.`J#` = sp.`J#`
-> WHERE j.city = s.city;
```

SNAME
Blake
Jones
Adams
Clark

4 rows in set (0.01 sec)

```
w1_99099_vaibhavi>
```

Q6) Display the Part name that is 'Red' in color, and the Total Quantity sold for that part.

```
ERROR 1146 (42S02): Table 'sp.sp' doesn't exist
w1_99099_vaibhavi>SELECT Pname, SUM(QTY)
-> FROM p
-> JOIN sp ON p.`P#` = sp.`P#`
-> WHERE p.Color = 'Red'
-> GROUP BY p.Pname;
+-----+-----+
| Pname | SUM(QTY) |
+-----+-----+
| Nut   |      1000 |
| Screw |      1300 |
| Cog   |      1300 |
+-----+-----+
3 rows in set (0.00 sec)

w1_99099_vaibhavi>
```

Q7) Display all the Quantity sold by Suppliers with the Status = 20

```
w1_99099_vaibhavi>SELECT s.SNAME, SUM(sp.QTY)
-> FROM s
-> JOIN sp ON s.`S#` = sp.`S#`
-> WHERE s.STATUS = 20
-> GROUP BY s.SNAME;
+-----+-----+
| SNAME | SUM(sp.QTY) |
+-----+-----+
| Smith |          900 |
| Clark |          600 |
+-----+-----+
2 rows in set (0.00 sec)

w1_99099_vaibhavi>
```

Q8) Display all the Parts and their Total Quantity for the parts with the Weight > 14.

```
w1_99099_vaibhavi>SELECT Pname, weight, SUM(sp.QTY)
-> FROM p
-> JOIN sp ON p.`P#` = sp.`P#`
-> WHERE p.weight > 14
-> GROUP BY p.`P#`, p.Pname;
+-----+-----+-----+
| Pname | weight | SUM(sp.QTY) |
+-----+-----+-----+
| Screw |      17 |          3500 |
| Cog   |      19 |          1300 |
| Bolt  |      17 |           300 |
+-----+-----+-----+
3 rows in set (0.00 sec)

w1_99099_vaibhavi>
```

Q9) Display all the Job names and City, which has bought more than 500 Parts.

```
w1_99099_vaibhavi>SELECT j.Jname, j.City, SUM(sp.QTY)
-> FROM j
-> JOIN sp ON j.`J#` = sp.`J#`
-> GROUP BY j.`J#`, j.Jname, j.city
-> HAVING SUM(sp.QTY) > 500;
```

Jname	City	SUM(sp.QTY)
Sorter	Paris	800
Console	Athens	3300
Punch	Rome	1200
Collator	London	1100
Tape	London	1200

5 rows in set (0.00 sec)

```
w1_99099_vaibhavi>
```

Q10) Display all the Part names and Quantity sold that have a Weight less than 15.

```
w1_99099_vaibhavi>SELECT p.Pname, SUM(sp.QTY) AS `sum(sp.QTY)`
-> FROM p
-> JOIN sp ON p.`p#` = sp.`p#`
-> WHERE p.weight < 15
-> GROUP BY p.`P#`, p.Pname;
```

Pname	sum(sp.QTY)
Nut	1000
Cam	1100
Screw	1300

3 rows in set (0.00 sec)

```
w1_99099_vaibhavi>
```

Q11) Display all the Suppliers with the same Status as the supplier, 'CLARK'.

```
w1_99099_vaibhavi>select sname, status
-> FROM s
-> WHERE STATUS = (
-> SELECT STATUS
-> FROM s
-> WHERE SNAME = 'Clark'
-> )
-> AND SNAME <> 'Clark';
+-----+-----+
| sname | status |
+-----+-----+
| Smith |      20 |
+-----+-----+
1 row in set (0.00 sec)

w1_99099_vaibhavi>
```

Q12) Display all the Parts which have more Weight than any “Red” parts.

```
w1_99099_vaibhavi>SELECT PNAME, WEIGHT, COLOR
-> FROM p
-> WHERE COLOR IN ('Red', 'Blue', 'Green')
-> AND WEIGHT >= 14
-> ORDER BY WEIGHT DESC, PNAME ASC;
+-----+-----+-----+
| PNAME | WEIGHT | COLOR |
+-----+-----+-----+
| Cog   |      19 | Red   |
| Bolt  |      17 | Green |
| Screw |      17 | Blue  |
| Screw |      14 | Red   |
+-----+-----+-----+
4 rows in set (0.00 sec)

w1_99099_vaibhavi>
```

Q13) Display all the Jobs going on in the same city as the job ‘TAPE’.


```
w1_99099_vaibhavi>SELECT *
-> FROM j
-> WHERE CITY = (Select city from j WHERE Jname = 'Tape');
+-----+-----+-----+
| J# | Jname   | City   |
+-----+-----+-----+
| J5 | Collator | London |
| J7 | Tape    | London |
+-----+-----+-----+
2 rows in set (0.00 sec)

w1_99099_vaibhavi>|
```

Q14) Display all the Parts with Weight less than any of the “Green” parts.

```
w1_99099_vaibhavi>SELECT Pname, Weight
-> FROM p
-> WHERE weight < ANY (
-> SELECT weight
-> FROM p
-> WHERE Color = 'Green');
+-----+-----+
| Pname | Weight |
+-----+-----+
| Nut   | 12     |
| Screw | 14     |
| Cam   | 12     |
+-----+-----+
3 rows in set (0.01 sec)

w1_99099_vaibhavi>|
```

Q15) Display the name of the Supplier who has sold the maximum Quantity (in one sale).

```
w1_99099_vaibhavi>SELECT s.SNAME, MAX(sp.QTY)
-> FROM s
-> JOIN sp ON s.`S#` = sp.`S#`
-> GROUP BY s.SNAME
-> ORDER BY MAX(sp.QTY) DESC
-> LIMIT 1;
```

```
+-----+-----+
| SNAME | MAX(sp.QTY) |
+-----+-----+
| Adams |          800 |
+-----+-----+
1 row in set (0.00 sec)
```

```
w1_99099_vaibhavi>
```

Q16) . Display the name of the Supplier who has sold the maximum overall Quantity (sum of Sales).

```
w1_99099_vaibhavi>select s.sname, sum(sp.qty)
-> FROM s
-> JOIN sp ON s.`S#` = sp.`S#`
-> GROUP BY s.SNAME
-> ORDER BY SUM(sp.QTY) DESC
-> LIMIT 1;
```

```
+-----+-----+
| sname | sum(sp.qty) |
+-----+-----+
| Jones |         3200 |
+-----+-----+
1 row in set (0.00 sec)
```